Sovereign cloud

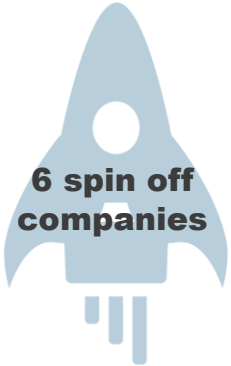
Eddy Truyen

DistriNet in a Nutshell

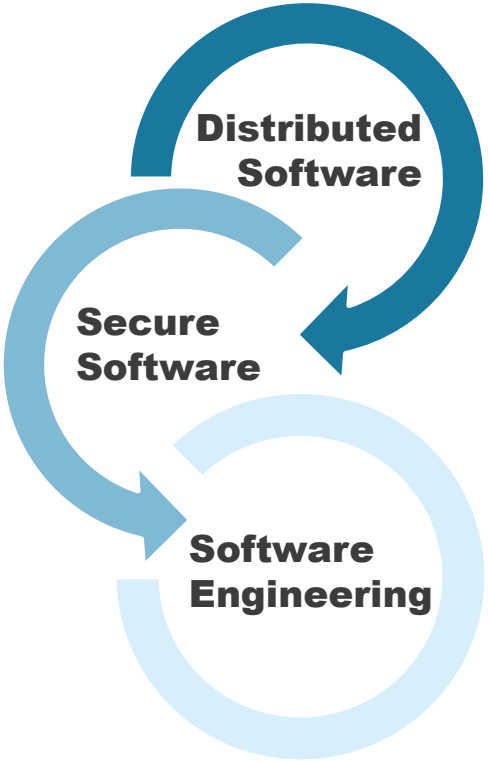
130+ people

19 professors
9 research managers
3 business office
12 postdocs
90+ PhD students

20+ years track record



6 spin off companies



30+ research projects

100 + industry collaborations

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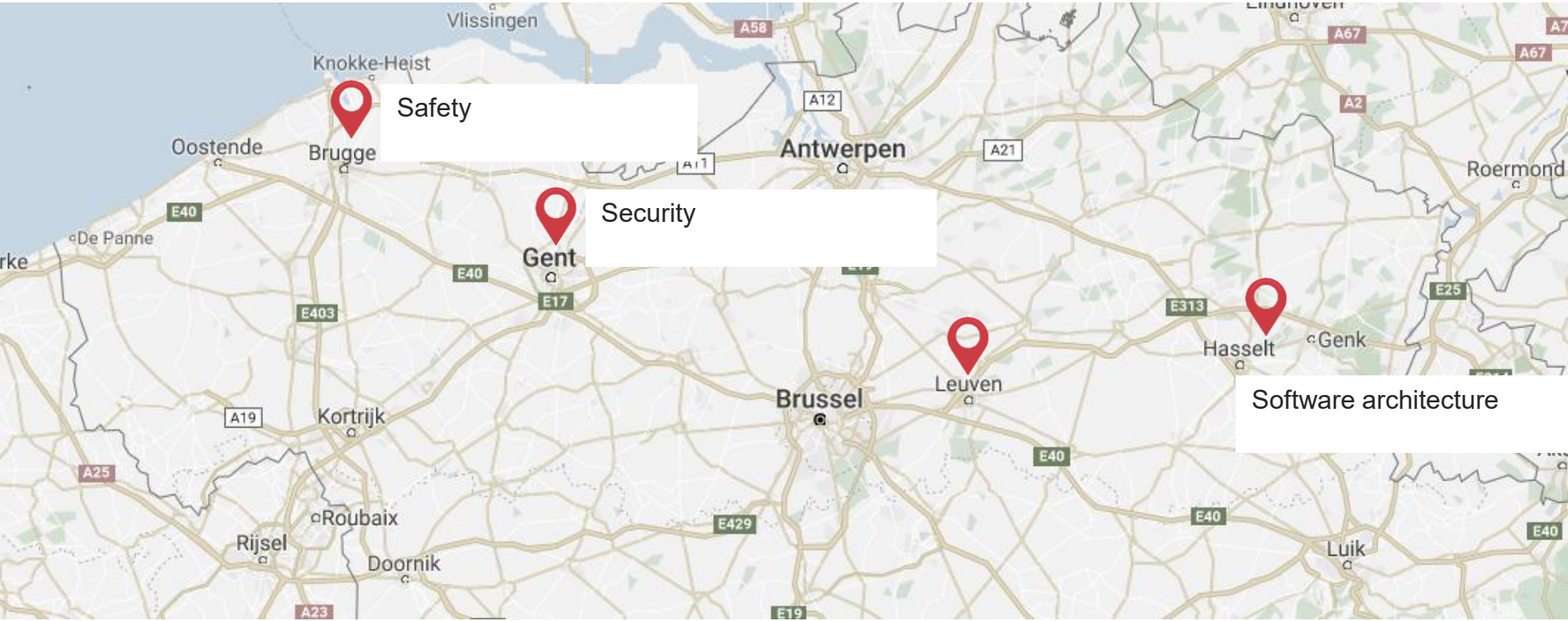
Present in Flanders



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Agenda

- **Analysis**
- Overview of research topics
- Discussion

European Sovereign cloud definition

in* Europe**: A cloud environment that ensures data remains in Europe without exposure to foreign access laws. ***Data sovereignty

for* Europe**: Compliant with legal, regulatory, and operational regulations of a specific jurisdiction. ***Operational sovereignty

by* Europe**. European cloud providers and open-source cloud stacks. ***Technical sovereignty

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Draghi report vs Now

Clouds

- **Headline of report[1]:**
  - Too late to turn dominance of hyperscalers
    - Sovereign cloud solutions = security and encryption
  - Public sector procurement
  - Outside of “sovereign” market segments: low barrier digital transatlantic market place
- **Current (perception of) reality**
  - US no longer reliable partner (trade wars, kill switch, AI-2027)

[1] Mario Draghi, The future of European competitiveness, Link

Strategic Differences in Digital Sovereignty

Risk management vs. “Buy European”

- **Cloud Sovereignty Framework (CSF)[1]:**
  - **Technical assessment tool specifically for Cloud services procurement**
- **Buy European Framework (BEF)[2]:**
  - **Proposed Regulation covering the entire strategic digital stack (AI, Hardware, Cloud) across the EU**

[1] European Commission, Cloud Sovereignty Framework, version 1.2.1 – Oct 2025, Link

[2] Eurostack Industry Initiative, A proposed framework for a “Buy European” Regulation of Strategic Digital Procurement -- 29 September 2025, Link

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Strategic Differences in Digital Sovereignty

Risk management vs. “Buy European”

- **CSF: Weighted Scoring**
  - Uses a Sovereignty Score (e.g., Supply Chain 20%, Legal 10%) to rank tenders alongside technical quality.

- **BEF: Pass/Fail Hierarchy**
  - Level 1 (Jurisdiction) acts as a mandatory gate. Non-EU parented entities are excluded before evaluation starts.

Design for Adaptability or Massively Invest?

- **Time is the bottleneck**
- **Technical: Alternatives do exist**
  - https://euro-stack.com/alternatives/
- **Full sovereignty impossible**: Microsoft Windows, Android, Apple
- **Social and economic perspective**
  - Social networks: users stick to what everybody uses
  - European payments:
    - Visa/Mastercard are very powerful
    - Digital Euro?

Chips, Clouds and AI

	KEY COUNTRIES	KEY FIRMS
Data and artificial intelligence	US, China	*OpenAI, Microsoft, Google, Meta, Anthropic, XAI, Amazon, Baidu, Tencent, Alibaba, DeepSeek*
Software	US, China, Germany	*Microsoft, Apple, Alphabet, Meta, Amazon, Salesforce, SAP, ByteDance, Tencent*
Cloud	US, China	*Amazon, Microsoft, Alphabet, Alibaba*
Internet of things & devices	US, China, Korea, Germany	*Amazon, Google, Apple, Samsung, Huawei, Bosch, Siemens, Xiaomi*
Networks	US, China, Europe, Japan	*Huawei, Nokia, Ericsson, ZTE, SpaceX, NEC*
Chips	Taiwan, Korea, US, Netherlands	*TSMC, Samsung, Intel, NVIDIA, AMD, ASML*
Raw materials, energy, and water	US, China, Russia	*Chinese government (through SOEs e.g., China Rare Earth Group), ExxonMobil, Gazprom*

F. Bria, P. Timmers and F. Gernone, “Eurostack – A European Alternative for Digital Sovereignty”, BertelmanssStifftung, Link

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Strengths of Europe

Edge computing and telecom

- Industry 4.0
- Near edge computing / fog computing
- The important yet dual role of telecom
  - Fragmented European market
  - National ex-ante regulations should be corrected
- Technical standards for edge computing as for roaming in the 90's
- Edge AI

Strengths of Europe

Europe's HPC supercomputers

- Europe's supercomputers are among the most powerful
- Scientific computing, simulations
- Batch jobs
- Specialized Container runtimes
- Recently also support for GenAI training[1]

[1] Angel M Beltre et al Experience Deploying Containerized GenAI Services at an HPC Center. In Proceedings of the SC '25 Workshops, 2025, Link

Strengths of Europe

RISC-V

- Digital Autonomomy for RISC-V in Europe
- RISER project
- TRISTAN &ISOLDE
- Open HW Foundation

[1] Angel M Beltre et al Experience Deploying Containerized GenAI Services at an HPC Center. In Proceedings of the SC '25 Workshops, 2025, Link

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High-level overview

From Political Debate to Engineering Discipline

1. Sovereign-by-Design

(Plan)

Proactively model sovereignty risks in architecture before code is written.

2. Secure Development

(Code, Build, Test)

Automatically test for compliance and vulnerabilities during coding and CI/CD.

3. Sovereign Operations

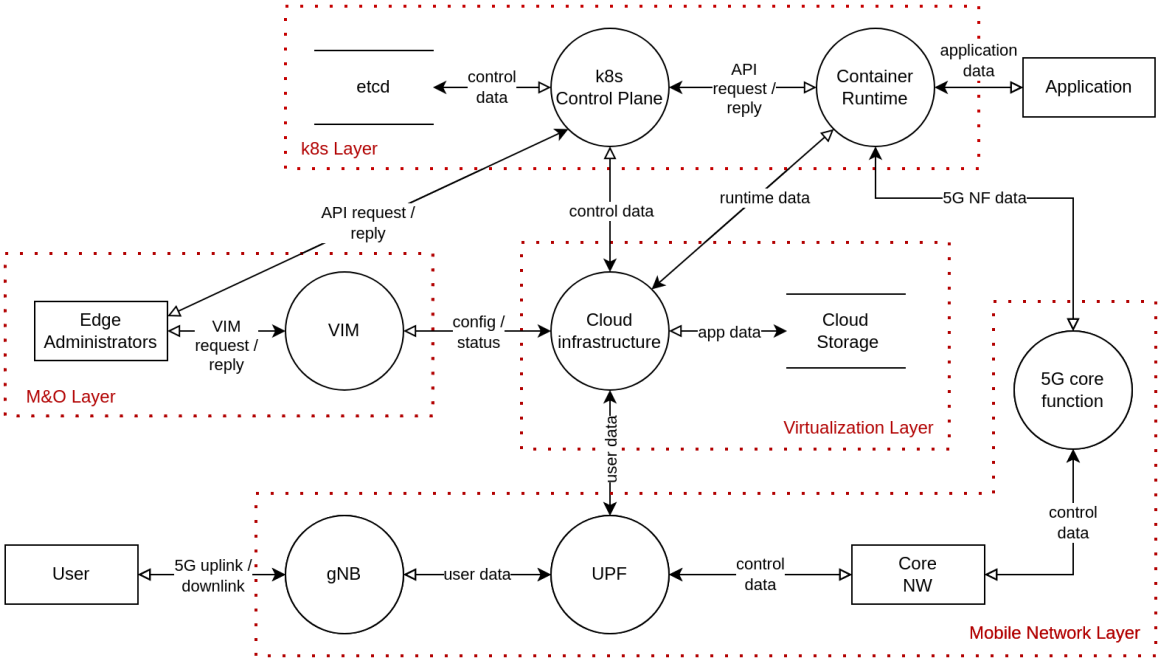
(Release, Deploy, Operate, Monitor)

Enforce policy and enable automatic orchestration in runtime environments.

Sovereign-by-design

Requirements analysis and architecture

- **Threat modeling**
  - Data Flow Diagram



[1] C. Baumann et al. “Automated threat analysis for K8s-based edge computing in 5G”, under review

Sovereign-by-design

Requirements analysis and architecture

- **Threat modeling**
  - Threat Elicitation and prioritization

Mnemonic: Title	Description, Prerequisites, Severity, Likelihood, & Risk
ElevPriv5G: Unauthorized access to 5G core functions	*Description:* If an attacker manages to compromise a pod running 5G core services (AMF, SMF, or UPF) they get full access to all user data handled by these functions. Moreover the attacker may obtain credentials allowing lateral movement into the 5G core network or user devices or to send malicious requests to other network functions. *Prerequisites:* The attacker needs to run code on an edge platform that also hosts 5G core functions and break their network isolation and authentication / authorization to gain access. *Severity: High; Likelihood: Low; Risk: 4*

[1] C. Baumann et al. “Automated threat analysis for K8s-based edge computing in 5G”, under review

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Sovereign-by-design

Requirements analysis and architecture

- **Threat modelling for digital sovereignty?**
  - Modeling the threats of consuming a European cloud provider
  - Modeling the threats faced by a kill switch victim
  - Risk of honest-but-curious cloud provider
  - Vendor lock-in risk

Sovereign-by-design

Other topics

- **Metrics for measuring digital sovereignty**
  - Measures if you are in control
- **Change Impact Analyis**
  - How does change in one microservice affect other microservices

Secure development

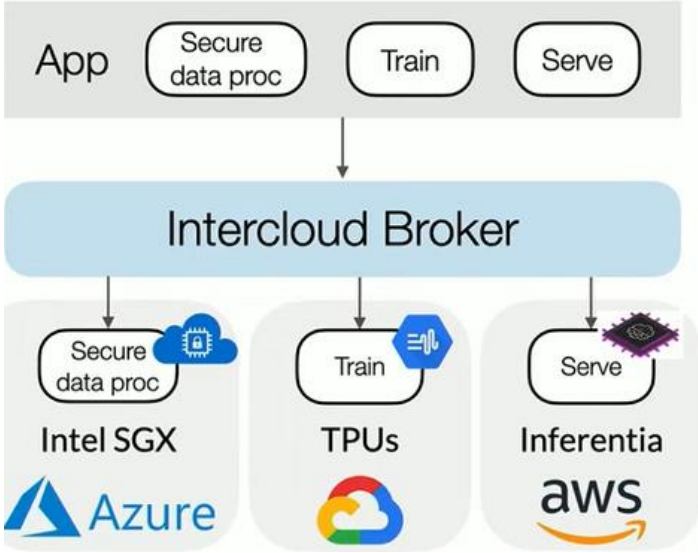
Topics

- **Supply chain sovereignty**
  - Complete and verifiable Software Bill of Materials (SBOM)
  - Audit right management
  - Degree of non-EU vendors assessed via dependency management
- **Technical sovereignty**
  - Standardized APIs for edge computing (K8s)

Sovereign operations

Hybrid cloud

Z. Yang, Z. Wu, M. Luo, W.-L. Chiang, R. Bhardwaj, W. Kwon, S. Zhuang, F. S. Luan, G. Mittal, S. Shenker, & I. Stoica, Sky Pilot: An Intercloud Broker for Sky Computing, Link



		proc	train	infer	egress	**Total**
Time	**Azure**	0.6	13.3	1.5	–	15.4
(hr.)	**Broker**	0.6	3.8 -71%	1.4 -7%	0.03	5.8 -62%
Cost	**Azure**	0.8	163	1.2	–	165
($)	**Broker**	0.8	32 -80%	0.5 -58%	0.1	33.4 -80%

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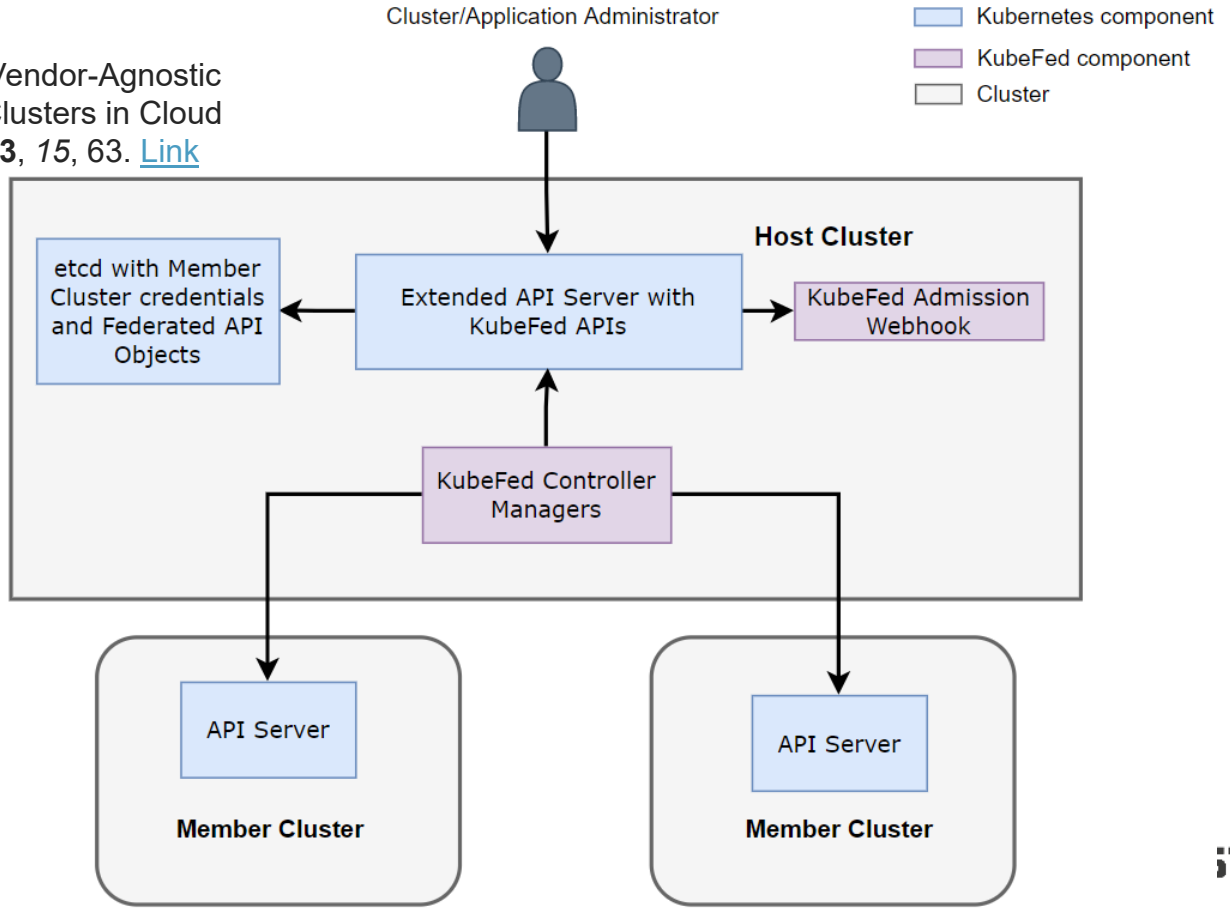
Vendor-agnostic reconfiguration of K8s clusters in hybrid cloud

Truyen, E.; Xie, H.; Joosen, W. Vendor-Agnostic Reconfiguration of Kubernetes Clusters in Cloud Federations. *Future Internet* **2023**, *15*, 63. Link



Vendor-agnostic reconfiguration of K8s clusters in hybrid cloud

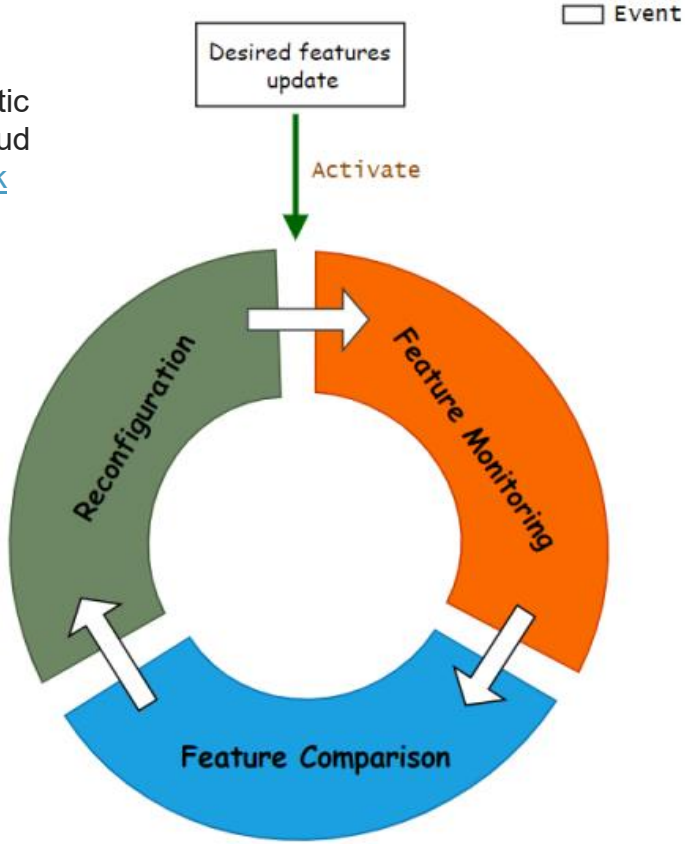
K8-as-a-service: feature incompatibilities

- K8s supports a wide range of strategies for authentication and authorization of users and worker nodes to control access to the Kubernetes API at master nodes
- Also authentication and authorization to control access to the kubelet API
- Audit (aka non-repudiation)

Type of access control	Existing triple A strategies of the k8s distribution	Azure AKS	AWS EKS	Google GKE
Authent. to master	X509 cert.	✓	✓	✓
	service account	✓	✓	✓
	bearer tokens	✓	✓	✓
Authent. to kubelet at port 10250	X509 cert.	✓	✓	✓
	bearer tokens		✓	✓
	no anonymous	✓	✓	✓
Author. to master	RBAC	✓	✓	✓
	ABAC			✓
	worker node authorization		✓	✓
Author. to kubelet	Webhook	✓	✓	✓
	no AlwaysAllow	✓	✓	✓
Audit			propr. alternative (CloudTrail)	✓

Operator for feature reconfiguration

Truyen, E.; Xie, H.; Joosen, W. Vendor-Agnostic Reconfiguration of Kubernetes Clusters in Cloud Federations. *Future Internet* **2023**, *15*, 63. Link



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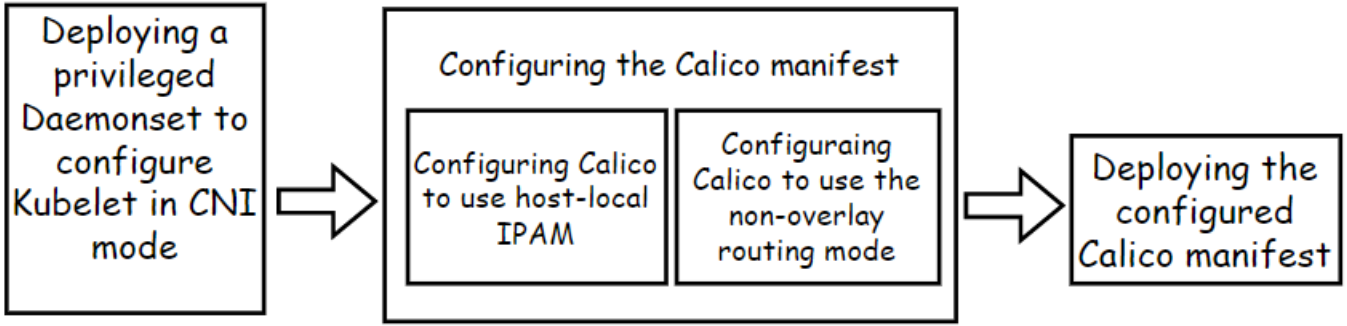
CNI reconfiguration using DaemonSet



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Sovereign operations

Hybrid cloud: the Uber case[1]

- Wants to migrate Spark / Presto workloads from on-premise to hybrid clouds
  - Objective: minimize egress costs, storage and network bandwidth costs for previously unseen jobs
  - Choice: replicate vs egress
  - Different strategies needed in function of migration progress: 30%, 50%, 70% migrated
  - Develop fully automated job placement and data replication solution
  - Findings:
    - Replicate Top 2.5% data by access traffic density
    - Light-weight access prediction is sufficient for effective job routing
    - Back-and-forth data movement is an optimization problem not a migration problem

[1] Z. Qiu et al. Moirai: Optimizing Placement of Data and Compute in Hybrid Clouds. 2025 In Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP '25). https://doi.org/10.1145/3731569.3764802

Sovereign operations

Edge AI

- Federated Learning + Differential Privacy / Homomorphic Encryption
- Network Intrusion detection
- Distributed Neural Networks
- AI-based orchestration
  - Microservice decomposition and chaining in NFV
  - Security-aware placement and migration of microservices

Sovereign operations

Other topics

- **Polyglot persistence**
  - Policy-based management of which data is managed by which database instance
- **Self-sovereign applications**
  - Wallet Identity and Decentralized Trust
- **Sustainability**
  - Energy-efficient Edge AI

Agenda

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Cloud providers: control full stack vs European modules

- Hyperscalers
  - Special chips hardware architecture design
    - Multi-tenant when possible or desired
  - Cloud services
- Uniting small players in Europe
  - SME’s and European Clouds
  - Letting servers of SMEs be managed by European CSPs
  - European modules

European cloud

- Europe has the best cloud servers and networks, but no services
  - Like selling wood, but customers want furnitures
- Key services that are weakly available among Europe's cloud players:
  - "S3: I got some push back on this. Folks say 'we have S3 at home', but seriously, the big cloud S3 variants beat the pants of whatever we have at home
  - AWS Simple Queuing Service (SQS)
  - IAM between services (this is the big one)
  - AWS Cognito equivalent
  - Rock solid managed Kubernetes
  - Managed database / RDS
  - Reliably sending email/notifications (!)
- => Opensource European cloud modules

https://berthub.eu/articles/posts/european-cloud-modules/

Sustainability

- Energy costs / Energy price
- Cooling / Water shortage
- Pollution
- Protests against datacenters and AI
  - https://www.theguardian.com/us-news/2026/feb/07/california-monterey-park-stop-datacenter-construction
  - https://time.com/7371825/trump-data-center-ai-backlash-ai-america-china/
  - https://www.wired.com/story/data-center-criticism-factories-supply-us/

What is needed to get there

- Pool of talents
  - “Cloud companies should pay real money (5 million euros/year), offer a glorious open source challenge, and real talent will arrive”
- But
  - “if we all try to build our individual AWS, we are going to fail. But cooperation is our strength here, so we should be able to make it happen.”
- “If governments, who think about more than the next quarter, show up as customers & moderate scale funders, and if legislation also creates a market for European clouds for sensitive/crucial data”.

https://berthub.eu/articles/posts/european-cloud-modules/

European Sovereign Clouds – Definition and Thought Leaders

Eddy Truyen, Tom Van Cutsem, Wouter Joosen

https://distrinet.cs.kuleuven.be/

European Sovereign cloud definition

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Sovereign clouds in Europe

- ***For*** and ***in*** Europe
  - T-systems (Germany)
    - operates Google distributed cloud (GDC) air-gapped
    - Open-source Sovereign cloud with Gardener (SAP)
  - Proximus-NXT (Luxemburg)
    - GDC air-gapped
  - Microsoft Cloud for Sovereignty (MCfS)
  - AWS European Sovereignty Cloud (end of 2025 launched)
- ***By*** Europe:
  - Sovereign cloud stack (SCS)
    - Based on K8s and OpenStack
  - OVH Cloud (France)
  - Hetzner (Germany)

Modern Air-Gapped Kubernetes

- Air-Gapped Environment Constraints
  - No internet access → all resources internal
  - Dependency management & deep dependency trees
  - PKI & certificate challenges → internal CAs
  - Limited CPU, memory, storage → careful planning
  - Security & verification → Sigstore, SLSA, SBOMs

Multi-cloud / Hybrid cloud

- Google Kubernetes Engine Multi-Cloud API
  - Run GKE cluster on AWS/Azure
  - Attach any CNCF compliant cluster to GKE project
  - The Multi-Cloud API is GKE-specific. It is not based on a standardized API
- Cluster API
  - Part of Management cluster with Cluster controller component

Thought leaders

Clingendael report is still up-to-date

- Clingendael report: Too late to act? Europe's quest for cloud sovereignty (2024)
- GAIA-X [1]
  - Strengths: agenda-setting, governance frameworks, community building in Europe.
  - Limitations: not yet delivering technical maturity, substitution capabilities, or SME-friendly solutions.

[1] Baur, A. (2025). European ambitions captured by American clouds: digital sovereignty through Gaia-X? *Information, Communication & Society*, 1–18.

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Clingendael report is still up-to-date

- Proposed sovereign cloud strategy in line with the Draghi report [1,2]:
  - “too late for the EU to try and develop systematic challengers to the major US cloud providers”
  - the EU should adopt policies “for collaboration between EU and non-EU cloud providers”.
  - “access to US hyperscalers’ latest cloud technologies while preserving encryption, security and ring-fenced services for trusted EU providers”.

[1] Ibid., pp. 20,30; M. Draghi, “The future of European competitiveness, Part B: In-depth analysis and recommendations”, pp.77, 82, 84.
[2] Michels, Johan David and Walden, Ian and Millard, Christopher, Storm Clouds are Building: Surveillance, Sovereignty, and State Interests (February 03, 2025).

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Draghi report

Roadblocks to innovation in Europe

- Static industrial structure (automotive)
- Weaknesses along the innovation cycle due to fragmented financing
- Not enough academic institutions achieving top excellence and the pipeline from innovation to commercialization is weak
- EU budget spending on R&I lacks scale and focus
- Barriers against scaling up young companies
- Digital bottlenecks (AI chip race)

IPCEI CIS

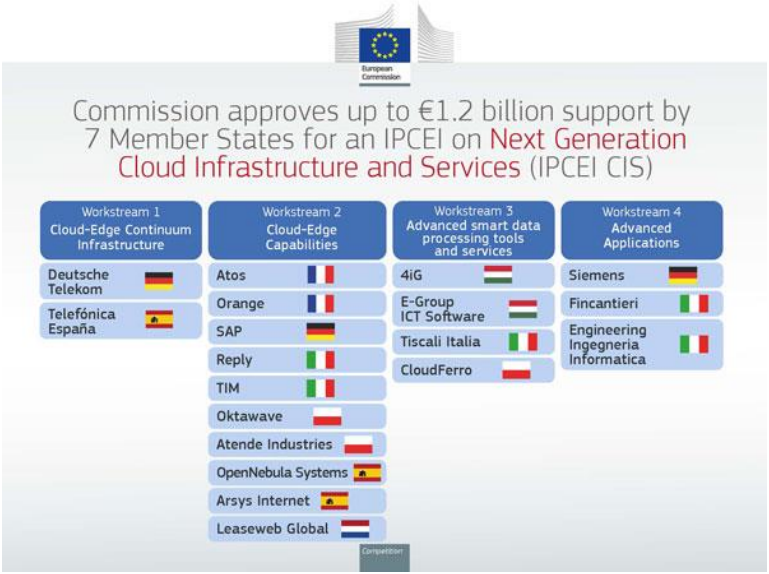
[IPCEI-CIS ↗]

[Fact8ra.AI]

[ONEnextgen ↗]

[8ra Initiative ↗]

[CISERO ↗]



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Bert Hubert’s blog

- Sovereign cloud [https://berthub.eu/articles/posts/hoe-soeverein-wil-je-zijn/]
  - Vertrouwelijkheid:
  - Wendbaarheid:
  - Weerbaarheid: Can we repair it ourselves?
  - Beschikbaarheid:
- What if Trump throws Europe out of the American Hyperscalers
  - Precedenten:
    - Hoofd Strafhof in den Haag toegang tot haar Outlook email ontzegd door Trump

Bert Hubert's blog

- Cloud Klootwijk: We can build it ourselves
  - Rijkscloud NL: Informatie- en communicatietechnologie (ICT) | Tweede Kamer der Staten-Generaal
- What is needed to get there:
  - "Cloud companies should pay real money (5 million euros/year), offer a glorious open source challenge, and real talent will arrive"
- But
  - "if we all try to build our individual AWS, we are going to fail. But cooperation is our strength here, so we should be able to make it happen."
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  - [https://berthub.eu/articles/posts/european-cloud-modules/]

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Academic thought leaders

Rajkumar Buyya – Intercloud (2011)
Concept of utility-oriented federation of clouds
Clouds as a 'utility grid'
Broker-based resource allocation
Focus: scaling, cost, utility

??

UC Berkeley – The Sky Above the Clouds (2022)
Cloud broker for compatibility matching
Digital sovereignty (and AI) as a motivating driver
"Sovereignty rules outpace cloud expansion—workloads must be restructured to use compliant cloud services."

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What has been published between?

Rajkumar Buyya – Intercloud (2011)
Concept of utility-oriented federation of clouds
Clouds as a 'utility grid'
Broker-based resource allocation
Focus: scaling, cost, utility

N. Khorasani *et al.*, "Cloud Broker: A Systematic Mapping Study," in *IEEE Transactions on Services Computing*, (2024) – **No Sovereignty-focused RQs**

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SkyPilot (NSDI 2023) (+100 citations)
Prototype of cloud broker for AI, by Berkeley

??

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Hybrid cloud and AI

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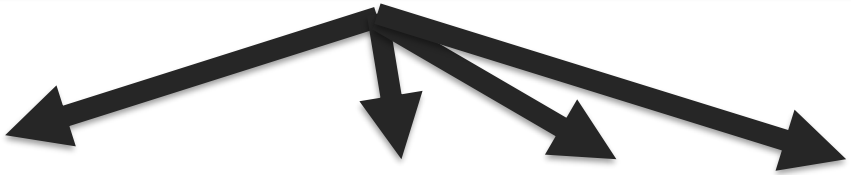

Qiu, Ziyue, et al. "Moirai: Optimizing Placement of Data and Compute in Hybrid Clouds." (SOSP 2025).

Mao, Ziming, et al. "Skyserve: Serving ai models across regions and clouds with spot instances." *Proceedings of the Twentieth European Conference on Computer Systems*. 2025.

Luo, Michael, et al. "Starburst: A Cost-aware Scheduler for Hybrid Cloud." *2024 USENIX Annual Technical Conference (USENIX ATC 24)*. 2024.

Chang, Tzu-Tao, and Shivaram Venkataraman. "Eva: Cost-Efficient Cloud-Based Cluster Scheduling." *Proceedings of the Twentieth European Conference on Computer Systems*. 2025.

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Sovereign cloud

“Sovereignty rules outpace cloud expansion—workloads must be restructured to use compliant cloud services.” (2 citations)

Kagermann H, Wahlster W. Ten Years of Industrie 4.0. *Sci*. 2022; 4(3):26

D. Tennenhouse, "Surprise-Inspired Networking," in *Computer*, vol. 56, no. 1, pp. 30-41, Jan. 2023

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Comparing European Sovereignty Frameworks

Cloud Sovereignty Framework (CSF) vs. Buy European Framework (BEF)

Detailed Comparative Analysis

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“Buy European” Procurement framework

- Level 1 (Mandatory Prerequisite / Pass-Fail):
  - Jurisdiction and governance: Ultimate accountability of EU law
- Level 2 (Core guarantees)
  - Technical Sovereignty: Freedom from technological lock-in
  - Operational Sovereignty: Control over the entire service delivery chain
  - Data Sovereignty: Verifiable protection of all data
- Level 3 (Key differentiator)
  - Economic Sovereignty: Net contribution to European economy

Eurostack Industry Initiative, A proposed framework for a “Buy European” Regulation of Strategic Digital Procurement

Cloud Sovereignty framework

- EU-defined Sovereignty Levels (SEALS 0 – 4)
  - 0: No Sovereignty
  - 1: Jurisdictional
  - 2: Data
  - 3: Digital
  - 4: Full: Technology and operations under complete EU control, subject only to EU law, with no critical non-EU dependencies.

1. Mapping of Common Criteria

Dimension	Cloud Sovereignty (CSF)	Buy European (BEF)
Legal & Jurisdictional	SOV-2: Immunity from foreign laws (e.g. US CLOUD Act)	Dim I: Ultimate Parent Entity must be in EU/EEA
Data Control	SOV-3: Data & AI residency and processing within EU	Dim IV: 'Bright-line' rule for exclusive EU residency
Operational Control	SOV-4: Control over service delivery and operations	Dim III: Exclusive European personnel/infrastructure
Technical Autonomy	SOV-6: Avoiding lock-in and ensuring portability	Dim II: Mandatory open standards / Open Source
Supply Chain	SOV-5: 20% weight on resilience against non-EU deps	Crit 3.3: Auditable strategy to mitigate foreign deps

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2. Strategic Differences: Scope

- **Cloud Sovereignty Framework (CSF):**
  - **Technical assessment tool specifically for Cloud services procurement.**
- **Buy European Framework (BEF):**
  - **Proposed Regulation covering the entire strategic digital stack (AI, Hardware, Cloud) across the EU.**

3. Assessment Methodology

- **CSF: Weighted Scoring**
- • Uses a Sovereignty Score (e.g., Supply Chain 20%, Legal 10%) to rank tenders alongside technical quality.

- **BEF: Pass/Fail Hierarchy**
- • Level 1 (Jurisdiction) acts as a mandatory gate. Non-EU parented entities are excluded before evaluation starts.

4. Unique Dimensions

- **CSF: Environmental Sustainability (SOV-8)**
- • Includes a specific 5% weight for environmental impact as a sovereignty objective.

- **BEF: Economic Sovereignty (Dim V)**
- • Evaluates R&D investment and contribution to the European ecosystem to prevent 'Sovereignty Washing'.

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Summary: Tactical vs. Structural

- CSF is a tool for evaluating risk in cloud tenders TODAY.
- BEF is a push for a structural change in EU law to favor European-controlled companies by DEFAULT.